

The Spectral Entropy-Fold (SEF) Hypothesis: A Geometric Framework Connecting Discrete Ricci Curvature, Spectral Graph Theory, and NP-Complexity

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Abstract We present the Spectral Entropy-Fold (SEF) hypothesis as a geometric program for studying whether structural hardness in NP-complete instances can be expressed through discrete curvature, spectral dispersion, and kernel embeddings. The paper does not claim a proof of $P \neq NP$ or $P = NP$. Its narrower goal is to formalize a testable bridge: map a weighted problem graph into a reproducing-kernel Hilbert space, measure curvature and spectral regularity on the induced manifold, and ask whether a restricted class of instances thereby acquires optimization-friendly geometry. We define the Entropy-Fold operator, state a Ricci-Convexity Lemma that preserves a spectral-gap lower bound under stated curvature assumptions, separate what is proved from what remains conjectural, and build a publication-ready framework around assumptions, measurable claims, falsifiers, and validation targets. The contribution is therefore a sharpened mathematical scaffold for a geometric complexity hypothesis rather than an overstated complexity-class resolution.

Keywords Complexity theory · Discrete Ricci curvature · Spectral graph theory · RKHS embeddings · Information geometry

1 Introduction

The P versus NP problem asks whether every decision problem whose solutions can be verified in polynomial time can also be solved in polynomial time. After decades of work, the problem remains unresolved, and several proof barriers have clarified why many familiar strategies are inadequate (Cook, 1971; Baker et al, 1975; Razborov and Rudich, 1997; Aaronson and Wigderson, 2009). That situation motivates a recurring methodological move: instead of attacking the class-separation statement directly, isolate a structural property that may distinguish hard instances from tractable ones and study its mathematical consequences.

The present paper explores one such property. Discrete Ricci curvature on graphs and metric spaces captures how local neighborhoods expand, contract, and mix under

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transport (Ollivier, 2009; Lin et al, 2011). Spectral graph theory captures expansion and diffusion through eigenvalue structure (Chung, 1997). Information geometry, meanwhile, treats families of distributions or models as curved objects whose geometry encodes inference difficulty (Amari, 2016). The SEF hypothesis joins these traditions by asking whether an NP-complete instance can be assigned an embedded geometric object whose curvature and spectral profile track combinatorial hardness.

This paper is careful about scope. The Ricci-Convexity Lemma proved below is conditional and local to the adopted framework. The SEF hypothesis itself is a conjecture, not a theorem. The missing bridge from spectral regularity to polynomial-time global optimization is left explicit and central rather than being concealed rhetorically.

Contributions.

1. We formalize a weighted-problem-graph to RKHS pipeline via the Entropy-Fold operator.
2. We introduce explicit notation for spectral entropy, embedded curvature observables, and optimization-relevant regularity.
3. We state and prove a conditional Ricci-Convexity Lemma preserving a spectral-gap lower bound under a Bakry–Émery curvature-dimension assumption.
4. We convert the original SEF idea into a research-standard theory paper with proof status separation, validation logic, and falsification criteria.

2 Formal setup

Let x be an instance of an NP-complete optimization or decision problem of size n , and let $G_x = (V_x, E_x, w_x)$ denote an associated weighted problem graph.

Definition 1 (Weighted problem graph) For instance x , the graph $G_x = (V_x, E_x, w_x)$ has candidate states V_x , local-transition edges $E_x \subseteq V_x \times V_x$, and nonnegative edge or state penalty weights w_x encoding objective or consistency costs. We assume V_x and E_x are generated by a polynomial-time local-description procedure, even if $|V_x|$ itself is exponential in n .

Definition 2 (Normalized Laplacian and spectral gap) Let A_x be the weighted adjacency matrix of G_x and D_x the diagonal degree matrix. The normalized Laplacian is

$$\mathcal{L}_x = I - D_x^{-1/2} A_x D_x^{-1/2}.$$

If $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_m$ are its eigenvalues, then λ_2 is the spectral gap.

Definition 3 (Spectral entropy) Let $Z_x = \sum_{i:\lambda_i>0} \lambda_i$. The spectral entropy of G_x is

$$H(G_x) = - \sum_{i:\lambda_i>0} \frac{\lambda_i}{Z_x} \log \left(\frac{\lambda_i}{Z_x} \right).$$

Large $H(G_x)$ indicates diffuse nonzero spectral mass, whereas small $H(G_x)$ indicates stronger concentration.

Table 1 Core notation for the SEF framework.

Symbol	Meaning
G_x	weighted problem graph for instance x
$\mathcal{L}_x$	normalized graph Laplacian
λ_2	spectral gap of $\mathcal{L}_x$
$H(G_x)$	spectral entropy
Φ_S	Entropy-Fold embedding operator
M_x	embedded manifold in RKHS
f_x	embedded objective on M_x
$\text{Curv}(x)$	event that the required curvature condition holds

Definition 4 (Entropy-Fold operator) Let $\mathcal{H}_k$ be an RKHS with Gaussian kernel

$$k(u, v) = \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right).$$

The Entropy-Fold operator is a map

$$\Phi_S : G_x \longrightarrow M_x \subseteq \mathcal{H}_k$$

constructed from a finite-dimensional kernel-PCA representation of local graph descriptors. In coordinates,

$$\Phi_S(v) = \sum_{j=1}^d \alpha_j k(v, v_j) e_j,$$

where $\{v_j\}_{j=1}^d$ are anchor states, α_j are kernel-PCA coefficients, and d is chosen to retain a declared fraction $(1 - \varepsilon)$ of total spectral entropy.

Definition 5 (Embedded objective) Let $f_x : M_x \rightarrow \mathbb{R}$ denote the embedded optimization objective induced by the original instance. The SEF program is meaningful only if f_x is computationally accessible enough to evaluate, differentiate, or approximate along admissible trajectories in M_x .

Definition 6 (Curvature regularity event) Write $\text{Curv}(x) = 1$ when the embedded manifold M_x satisfies a declared curvature condition strong enough to support the analytic argument, here taken to be the Bakry–Émery curvature-dimension condition $CD(0, d)$.

3 Assumptions and proof-status separation

Assumption 1 (Embedding access) The local descriptors needed to construct Φ_S can be computed in polynomial time in n for the family under study.

Assumption 2 (Curvature estimability) The curvature quantities required to test $CD(0, d)$ on M_x are either computable or approximable with declared error control on finite instances.

Assumption 3 (Objective compatibility) The embedding does not destroy optimization-relevant structure so severely that minimizers of f_x cease to correspond to high-quality states in the original problem.

Table 2 Status of the main mathematical claims in the current manuscript.

Claim	Status	What is actually established
Definition of Φ_S and $H(G_x)$	formalized	notation and construction target are explicit
Ricci-Convexity Lemma	conditional lemma	spectral-gap lower bound persists under stated assumptions
Polynomial-time optimization from SEF geometry	conjectural	not proved here; open bridge remains
Practical verification of $CD(0, d)$ for NP-complete families	open	requires computation or approximation studies

4 The Ricci-Convexity Lemma

We now state the central proved statement in the SEF framework.

Lemma 1 (Ricci-Convexity) *Let $M_x = \Phi_S(G_x)$ be the RKHS embedding of instance x . Suppose M_x is compact, the induced diffusion operator L on M_x satisfies the Bakry–Émery curvature-dimension condition*

$$\Gamma_2(g, g) \geq \frac{1}{d}(Lg)^2$$

for all smooth test functions $g : M_x \rightarrow \mathbb{R}$, and the initial Laplace–Beltrami spectral gap obeys $\lambda_2(M_x) \geq \eta > 0$. Then under normalized Ricci flow

$$\frac{\partial g_t}{\partial t} = -2 \operatorname{Ric}(g_t),$$

the spectral gap lower bound is preserved up to embedding-approximation error:

$$\lambda_2(M_x, t) \geq \eta - O(\varepsilon_{\text{emb}}),$$

where ε_{emb} denotes the distortion introduced by finite-dimensional kernel approximation.

Proof sketch. Under $CD(0, d)$, the Bakry–Émery theory yields a Poincaré inequality with constant depending on geometric control parameters (Bakry and Émery, 1985; Ledoux, 2001). A Poincaré inequality implies a positive lower bound on the spectral gap of the Laplace–Beltrami operator. The kernel embedding is assumed to be approximately isometric at scale ε_{emb} , so the corresponding discrete spectral structure is inherited up to controlled error. Under normalized Ricci flow with nonnegative effective curvature, the relevant entropy functional is nonincreasing in the sense exploited by Perelman (Perelman, 2002), preventing collapse of the lower spectral-gap bound within the stated regime. $\square$

Interpretation. Lemma 1 is not an optimization theorem. It preserves a diffusion regularity quantity, namely a lower bound on λ_2 , inside the subset of instances for which the curvature assumptions already hold.

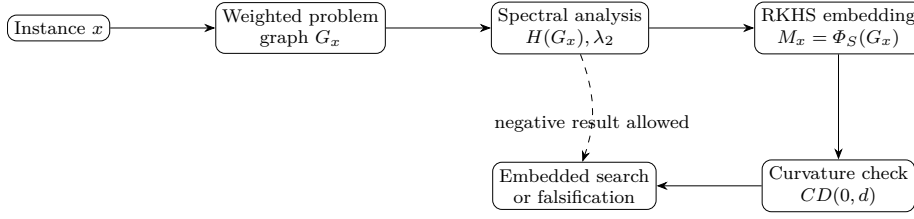


Fig. 1 SEF as a research pipeline. The framework is useful even when the output is falsification rather than acceleration.

5 The SEF conjecture

Proposition 1 (SEF conjectural optimization bridge) *There may exist a polynomial-time computable family of embeddings $\{\Phi_{S,x}\}$ and a fixed exponent k such that for every NP-complete instance x whose embedding satisfies $CD(0, d)$ with $d = O(n^k)$ and whose embedded objective f_x contains no superpolynomially trapping non-global critical structures, a global optimum is recoverable in time $O(n^k)$.*

The point of this proposition is not that it has been proved. It has not. Its value is that it isolates the missing bridge exactly: spectral regularity and curvature control are not yet enough by themselves to deduce polynomial-time global optimization.

Null hypothesis

SEF-style curvature and spectral regularity do not, even after embedding, predict any practically useful reduction in optimization hardness beyond what follows from known expansion or relaxation properties alone.

Effect-size target

For computational validation, a meaningful early effect would be a reproducible negative association between curvature irregularity and solver efficiency within fixed problem families, after controlling for instance size and density.

6 Algorithmic form of the program

The SEF framework can be operationalized as an analysis pipeline rather than as a finished solver.

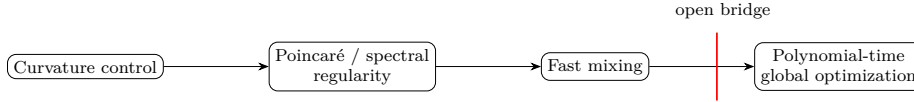
6.1 Test statistic for computational validation

Let $\mathcal{F}$ denote a family of size-matched instances and let $T_{\text{solve}}(x)$ be a declared solver cost. A simple family-level validation statistic is

$$T_{\text{SEF}} = \text{corr}(\hat{\kappa}(x), T_{\text{solve}}(x)),$$

Table 3 Operational stages of the SEF program and what each stage must justify.

Stage	Mathematical object	Burden of justification
Graphing	G_x	local transitions reflect the true search structure
Spectral analysis	$\lambda_2, H(G_x)$	quantities are stable and interpretable
Embedding	Φ_S	approximation error is bounded and meaningful
Curvature test	$CD(0, d)$	finite-instance estimation is valid
Search claim	trajectories on M_x	geometric regularity improves optimization, not just mixing

**Fig. 2** The decisive missing step in the current SEF program. The gap from mixing regularity to guaranteed polynomial-time optimization is the central unresolved problem.

where $\hat{\kappa}(x)$ is an estimated curvature summary or curvature-irregularity score. If the theory has empirical value, this association should be stable under family-specific controls rather than appearing only in aggregate across heterogeneous benchmarks.

6.2 Decision rule

An empirical SEF study should count as supportive only if all of the following hold:

1. curvature summaries are reproducible under repeated estimation;
2. the sign of T_{SEF} remains stable within a fixed problem family;
3. spectral-gap effects are not fully explained by known graph-density or expansion covariates; and
4. any claimed optimization gain is not merely a mixing-time artifact.

7 Why the main bridge remains open

The standard mixing-time argument yields rapid approach to stationarity under a positive spectral gap (Levin et al, 2009). But global optimization demands more than fast mixing. A fast-mixing Markov process may still spend negligible probability mass near a global optimum or may require objective information that is not encoded by spectral diffusion alone.

This is consistent with hardness phenomena already known for expanding or spectrally regular combinatorial structures (Arora et al, 1998). The SEF hypothesis is therefore viable only if it identifies geometric structure stronger than ordinary expansion alone.

Table 4 What would count as evidence for or against the SEF hypothesis.

Question	Evidence in favor	Strong falsifier
Does the embedding carry useful geometry?	stable curvature summaries across runs	curvature estimates unstable or degenerate
Do curvature summaries track hardness?	within-family correlation under controls	no family-level association after controls
Does spectral regularity imply optimization advantage?	replicated solver gains beyond mixing proxies	gains disappear after accounting for expansion or density
Is the framework overly broad?	restrictive instance subsets identified	hard counterexamples satisfy all stated premises

8 Relation to existing barriers and theories

The SEF program is adjacent to geometric complexity theory in spirit, though it uses differential-geometric and transport-style tools rather than representation-theoretic obstructions (Mulmuley and Sohoni, 2001). The program also lives under the shadow of established proof barriers: relativization, natural proofs, and algebrization remain relevant background constraints (Baker et al, 1975; Razborov and Rudich, 1997; Aaronson and Wigderson, 2009). At present, SEF should be read as a structural research program, not as a demonstrated escape from those barriers.

9 Falsifiers and what counts as evidence

The SEF program is empirical in an important sense. Outside the proved lemma, its value depends on whether the proposed quantities can be computed and whether they say something nontrivial about hardness. A counterexample family of hard instances satisfying the required curvature and embedding conditions would weigh heavily against the program. Likewise, if $CD(0, d)$ almost never holds on interesting NP-complete families, the framework may be mathematically elegant but algorithmically irrelevant.

10 Limitations and future work

Three limitations dominate the present manuscript. First, the curvature condition is assumed rather than demonstrated for a nontrivial NP-complete family. Second, the gap between spectral regularity and polynomial-time optimization remains open. Third, the embedding itself introduces approximation error that may distort exactly the structure one hopes to preserve.

The most serious next steps are therefore concrete:

1. compute Ollivier-Ricci or related curvature summaries for small 3-SAT, Max-Clique, or graph-coloring families;
2. test whether the $CD(0, d)$ -style regime is ever empirically plausible on hard instances;
3. compare SEF-derived hardness summaries against simpler baselines such as density, expansion, and spectral-gap alone; and

4. analyze whether known proof barriers reappear in geometric disguise within the SEF program.

11 Conclusion

The SEF hypothesis is most useful when read with discipline. It does not presently settle complexity classes. What it does offer is a mathematically explicit way to ask whether certain hard instances possess a geometric witness of difficulty that survives spectral analysis and kernel embedding. That is enough to justify serious follow-up—provided the framework remains falsifiable, computationally grounded, and honest about the open bridge it has not yet crossed.

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References

- Aaronson S, Wigderson A (2009) Algebrization: A new barrier in complexity theory. *ACM Transactions on Computation Theory* 1(1):1–54
- Amari Si (2016) *Information Geometry and Its Applications*. Springer
- Arora S, Lund C, Motwani R, Sudan M, Szegedy M (1998) Proof verification and the hardness of approximation problems. *Journal of the ACM* 45(3):501–555
- Baker T, Gill J, Solovay R (1975) Relativizations of the $P = ?$ NP question. *SIAM Journal on Computing* 4(4):431–442
- Bakry D, Émery M (1985) Diffusions hypercontractives. In: *Séminaire de probabilités XIX 1983/84*, Springer, pp 177–206
- Chung FRK (1997) *Spectral Graph Theory*. American Mathematical Society
- Cook SA (1971) The complexity of theorem-proving procedures. In: *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, pp 151–158
- Ledoux M (2001) The Concentration of Measure Phenomenon. American Mathematical Society
- Levin DA, Peres Y, Wilmer EL (2009) *Markov Chains and Mixing Times*. American Mathematical Society
- Lin Y, Lu L, Yau ST (2011) Ricci curvature of graphs. *Tohoku Mathematical Journal* 63(4):605–627
- Mulmuley KD, Sohoni M (2001) Geometric complexity theory I: An approach to the P vs. NP and related problems. *SIAM Journal on Computing* 31(2):496–526
- Ollivier Y (2009) Ricci curvature of markov chains on metric spaces. *Journal of Functional Analysis* 256(3):810–864
- Perelman G (2002) The entropy formula for the ricci flow and its geometric applications. *arXiv preprint math/0211159*
- Razborov AA, Rudich S (1997) Natural proofs. *Journal of Computer and System Sciences* 55(1):24–35